

Practical 1: Create a Free-Tier AWS Account and Explore AWS Console

Aim

To create an AWS Free Tier account and explore the AWS Management Console to understand the services offered by Amazon Web Services.

Procedure / Steps

Step 1: Open AWS Management Console

Open a web browser, search for *AWS Login Console*, and click on *AWS Management Console* from the search results.

Step 2: Start Account Creation

On the AWS homepage, click *Create an AWS Account* and choose to create a new AWS account.

Step 3: Enter Email and Account Details

Enter a valid email address and AWS account name. Click *Verify Email Address* and enter the verification code received via email.

Step 4: Create Password

Create and confirm a strong password following AWS security rules. Click *Continue*.

Step 5: Provide Contact Information

Select *Personal* account type. Enter full name, address, city, state, country, and phone number. Click *Continue*.

Step 6: Payment Information Verification

Enter debit/credit card details. AWS may charge a small temporary amount (approximately ₹2) for verification. No actual charges apply under the Free Tier. Click *Verify and Continue*.

Step 7: Phone Verification

Select verification method (SMS or Voice call). Enter mobile number, complete CAPTCHA, receive OTP, and verify.

Step 8: Select Support Plan

Choose *Basic Support Plan (Free)*, which provides 24×7 self-service access, AWS documentation, and forums at no additional cost. Click *Complete Sign Up*.

Step 9: Account Creation Confirmation

A confirmation message is displayed. Account activation may take a few minutes. Click *Sign in to the AWS Management Console*.

Step 10: Login to AWS Console

Enter registered email ID and password to log in successfully.

Step 11: Explore AWS Management Console

Observe the dashboard and explore services such as EC2, S3, IAM, VPC, and CloudWatch. Use the search bar, region selector, and recent services section.

Result

An AWS Free Tier account was successfully created, and the AWS Management Console was explored.

Conclusion

AWS Free Tier provides an excellent platform for beginners to learn cloud computing by offering hands-on exposure to compute, storage, security, and monitoring services.

Practical 3: Create and Configure an EC2 Instance

Aim

To create and configure an Amazon EC2 instance using the AWS Management Console.

Requirements

AWS Free Tier account, internet connection, web browser, and basic knowledge of cloud computing.

Theory

Amazon EC2 provides scalable virtual servers known as instances, allowing users to configure CPU, memory, storage, and networking without physical hardware.

Procedure / Steps

Step 1: Login to AWS Management Console

Open a web browser and log in using registered credentials.

Step 2: Open EC2 Service

Search for *EC2* in the console and open the EC2 Dashboard.

Step 3: Launch a New EC2 Instance

Click *Launch Instance*.

Step 4: Choose Amazon Machine Image (AMI)

Select *Amazon Linux 2 AMI* (Free Tier eligible).

Step 5: Choose Instance Type

Select *t2.micro* (1 vCPU, 1 GB RAM, Free Tier eligible).

Step 6: Configure Instance Details

Use default settings: Default VPC, Default Subnet, Auto-assign Public IP enabled.

Step 7: Add Storage

Keep default 8 GB EBS storage.

Step 8: Add Tags

Add a tag with Key: *Name* and Value: *MyEC2Instance*.

Step 9: Configure Security Group

Create a new security group and add inbound rules for SSH (Port 22) and HTTP (Port 80) from Anywhere.

Step 10: Create or Select Key Pair

Create a new key pair, download the `.pem` file securely, and launch the instance.

Step 11: Instance Launch Confirmation

Click *View Instances* after launch confirmation.

Step 12: Verify EC2 Instance Status

Ensure instance state is *Running* and note Instance ID, Public IPv4 address, and instance type.

Result

An Amazon EC2 instance was successfully created and configured.

Conclusion

Amazon EC2 enables flexible and scalable cloud infrastructure while providing hands-on experience in instance management and security configuration.

Practical 4: Store and Retrieve Files from an S3 Bucket**Aim**

To create an Amazon S3 bucket and store and retrieve files using the AWS Management Console.

Requirements

AWS Free Tier account, internet connection, web browser, and a sample file.

Theory

Amazon S3 is an object storage service that stores data as objects within buckets, offering high scalability, durability, and security.

Procedure / Steps**Step 1: Login to AWS Management Console**

Log in using valid credentials.

Step 2: Open Amazon S3 Service

Search for *S3* and open the S3 dashboard.

Step 3: Create an S3 Bucket

Click *Create Bucket*, enter a unique bucket name, and select a region.

Step 4: Configure Bucket Settings

Leave default settings unchanged and keep *Block all public access* enabled.

Step 5: Review and Create Bucket

Review configuration and click *Create Bucket*.

Step 6: Open the Created Bucket

Select the newly created bucket from the list.

Step 7: Upload Files

Click *Upload*, add files or folders, and start upload.

Step 8: Configure Upload Properties

Keep storage class as *Standard* and default permissions and encryption.

Step 9: Confirm File Upload

Verify upload status as *Succeeded*.

Step 10: Retrieve the Stored File

Select the file and click *Download*.

Step 11: Verify File Integrity

Open the downloaded file and verify its contents.

Result

An Amazon S3 bucket was created successfully, and files were uploaded and retrieved.

Conclusion

Amazon S3 provides secure and reliable cloud storage, giving users practical experience in managing cloud-based data.

Practical 5: Configure AWS Identity and Access Management (IAM) for a Multi-User Environment**Aim**

To create and configure AWS IAM users, groups, and policies to manage secure access to AWS resources in a multi-user environment.

Requirements

AWS Free Tier account, internet connection, and a web browser.

Theory

AWS Identity and Access Management (IAM) is a service that enables secure control of access to AWS resources. IAM allows administrators to create users, assign permissions, and manage access policies. It follows the principle of least privilege, ensuring users have only the permissions required to perform their tasks.

Procedure / Steps**Step 1: Login to AWS Management Console**

Open a web browser, go to the AWS Management Console, and log in using AWS root account credentials.

Step 2: Open IAM Service

In the AWS Console search bar, type *IAM* and select *Identity and Access Management (IAM)*. The IAM dashboard will open.

Step 3: Create an IAM User

Click *Users* from the left navigation panel and select *Add users*. Enter a user name (e.g., *IAMUser1*). Enable *AWS Management Console access*. Set a custom or auto-generated password and enable *Require password reset*. Click *Next*.

Step 4: Create an IAM Group

On the permissions page, click *Create group*. Enter a group name (e.g., *AdminGroup* or *DeveloperGroup*) and proceed.

Step 5: Attach Policies to the Group

Attach appropriate policies such as *AdministratorAccess* or *AmazonEC2FullAccess* to the group and click *Create group*.

Step 6: Add User to Group

Select the created group and add the IAM user to it. Click *Next*.

Step 7: Review and Create User

Review the user details and assigned permissions, then click *Create user*. The IAM user is created successfully.

Step 8: Download Login Credentials

Download the *.csv* file containing the IAM username, console login URL, and password. Store the credentials securely.

Step 9: Sign In Using IAM User

Open the IAM user login URL, enter the IAM username and password, reset the password when prompted, and log in successfully.

Step 10: Verify Permissions

Access permitted AWS services such as EC2. Attempt restricted services to verify permission limitations and confirm IAM policy enforcement.

Step 11: Enable Security Best Practices (Optional)

Enable Multi-Factor Authentication (MFA), use strong passwords, and avoid using the root account for daily operations.

Result

AWS IAM users and groups were successfully created and configured, and access permissions were managed using IAM policies.

Conclusion

AWS IAM provides secure access control for cloud resources. By configuring users, groups, and policies, a secure multi-user AWS environment can be implemented following industry best practices.

Practical 6: Configure Cloud Security Management Using Multi-Factor Authentication (MFA)

Aim

To configure Multi-Factor Authentication (MFA) in AWS in order to enhance cloud account security.

Requirements

AWS Free Tier account, mobile phone with Google Authenticator or similar MFA app, internet connection, and a web browser.

Theory

Multi-Factor Authentication (MFA) is a security mechanism that requires users to provide two or more authentication factors to access an account. In AWS, MFA adds an extra layer of protection by requiring both a password and a one-time code generated by an authenticator app. This prevents unauthorized access even if login credentials are compromised.

Procedure / Steps

Step 1: Login to AWS Management Console

Open a web browser, go to the AWS Management Console, and log in using root account or IAM user credentials.

Step 2: Open Security Credentials

Click on the account name at the top-right corner and select *Security Credentials* from the dropdown menu.

Step 3: Open MFA Section

Scroll down to the *Multi-Factor Authentication (MFA)* section and click *Assign MFA device*.

Step 4: Select MFA Device Type

Choose *Virtual MFA device* and click *Continue*.

Step 5: Install Authenticator Application

Install Google Authenticator or any compatible MFA application on your mobile phone and open the app.

Step 6: Scan QR Code

Scan the QR code displayed on the AWS Console using the authenticator app. The app will begin generating 6-digit time-based OTPs.

Step 7: Enter MFA Codes

Enter the first 6-digit code generated by the app, wait for the code to refresh, then enter the second 6-digit code. Click *Assign MFA*.

Step 8: MFA Configuration Confirmation

AWS displays a success message confirming that the MFA device has been successfully assigned.

Step 9: Sign Out and Test MFA

Log out of the AWS Console and log in again using username and password. Enter the MFA code when prompted to complete login.

Step 10: Verify MFA Protection

Confirm that login is not possible without the MFA code, ensuring the account is protected with two-factor authentication.

Result

Multi-Factor Authentication (MFA) was successfully configured and verified for the AWS account.

Conclusion

Enabling MFA significantly enhances cloud security by adding an additional authentication layer. MFA protects AWS accounts from unauthorized access and follows recommended security best practices.

Practical 8: Study and Implementation of Single Sign-On (SSO)

Aim

To study and implement Single Sign-On (SSO) using a third-party authentication provider.

Requirements

Internet connection, web browser, email ID, and an SSO provider account (Auth0 or Google Authentication).

Theory

Single Sign-On (SSO) is an authentication mechanism that allows users to access multiple applications using a single set of login credentials. Once authenticated, users do not need to log in again for each service. SSO improves user experience, reduces password fatigue, and enhances security by centralizing authentication through a trusted identity provider.

Procedure / Steps

Step 1: Open SSO Provider Website

Open a web browser, visit the SSO provider website (e.g., Auth0), and click *Sign Up* or *Create Account*.

Step 2: Create an SSO Account

Register using a valid email ID, verify the email address, and log in to the SSO provider dashboard.

Step 3: Create a New Application

From the dashboard, click *Create Application*, select *Regular Web Application*, and click *Create*.

Step 4: Configure Authentication Settings

Navigate to *Authentication* → *Social*, enable *Google* as the social login provider, and save the changes.

Step 5: Configure Connection Settings

Go to *Connections*, select *Social Connections*, and enable Google login for the created application.

Step 6: Test SSO Connection

Click *Try Connection*. A Google login window appears. Log in using Google credentials.

Step 7: Verify SSO Login

After successful login, the user is authenticated without requiring additional credentials for connected services.

Step 8: Observe SSO Behavior

Log out and log in again. Observe reduced login steps and seamless access using the same account.

Step 9: Validate Authentication Result

Successful login confirms correct SSO implementation, with user identity verified by the identity provider.

Result

Single Sign-On (SSO) was successfully implemented, allowing users to authenticate using a single login across multiple services.

Conclusion

SSO simplifies authentication by eliminating repeated logins and enhances security through centralized identity management. It is widely used in enterprise and cloud-based environments.

Advantages of SSO

- Reduces password fatigue
- Improves user experience
- Provides centralized authentication
- Enhances security
- Saves time

Practical 9: Create CloudWatch Metrics and Alarms for an EC2 Instance

Aim

To monitor an Amazon EC2 instance using AWS CloudWatch and create alarms based on performance metrics.

Requirements

AWS Free Tier account, running EC2 instance, internet connection, and a web browser.

Theory

Amazon CloudWatch is a monitoring and observability service that provides data and actionable insights for AWS resources. It collects metrics such as CPU utilization, network traffic, and disk activity, and triggers alarms when predefined thresholds are crossed. CloudWatch helps maintain performance, availability, and reliability of cloud resources.

Procedure / Steps

Step 1: Login to AWS Management Console

Open a web browser and log in to the AWS Management Console.

Step 2: Open CloudWatch Service

Search for *CloudWatch* in the AWS Console and open the CloudWatch Dashboard.

Step 3: Navigate to Metrics

Click *Metrics* from the left panel, select *EC2*, then choose *Per-Instance Metrics* and select the required Instance ID.

Step 4: View EC2 Metrics

Observe metrics such as CPU Utilization, Network In/Out, and Disk Read/Write. These metrics are updated automatically.

Step 5: Generate Load on EC2 Instance

Connect to the EC2 instance using SSH, install a stress tool if required, and generate load to increase CPU usage.

Step 6: Create CloudWatch Alarm

Go to *Graphed Metrics*, click *Create Alarm*, and select the *CPU Utilization* metric.

Step 7: Configure Alarm Conditions

Set the condition to *Greater than threshold* and set the threshold value to *80% CPU utilization*. Click *Next*.

Step 8: Configure Notification

Create a new SNS topic, enter an email address for notification, and confirm the email subscription.

Step 9: Name and Create Alarm

Provide an alarm name and description, review the configuration, and click *Create Alarm*.

Step 10: Verify Alarm Status

Observe the alarm state change when the threshold is crossed and verify that an email notification is received.

Result

CloudWatch successfully monitored EC2 instance metrics and generated alerts when the defined threshold was exceeded.

Conclusion

CloudWatch enables real-time monitoring and alerting, helping maintain performance and reliability of AWS resources.

Practical 10: Set Limits for AWS Resources Using Service Quotas**Aim**

To view and manage AWS service limits using the Service Quotas feature.

Requirements

AWS Free Tier account, internet connection, and a web browser.

Theory

AWS Service Quotas allows users to view, manage, and request increases for service limits. These limits prevent accidental overuse of resources and help control costs. Each AWS service has predefined quotas that can be adjusted based on requirements.

Procedure / Steps

Step 1: Login to AWS Management Console

Open a web browser and log in to the AWS Management Console.

Step 2: Open Service Quotas

Search for *Service Quotas* and open the Service Quotas Dashboard.

Step 3: Select AWS Service

Click *AWS Services* and select the required service (e.g., EC2).

Step 4: View Service Quotas

View default quotas such as EC2 instances, Elastic IPs, and vCPUs. Each quota displays the current limit and usage.

Step 5: Select a Quota

Click on a specific quota (e.g., EC2 vCPUs) to view detailed information.

Step 6: Request Quota Increase

Click *Request Quota Increase*, enter the desired new limit, and submit the request.

Step 7: Track Request Status

Monitor the quota request status while AWS reviews and approves it based on eligibility.

Step 8: Verify Updated Quota

Once approved, verify that the updated quota is reflected and resources can be used within the new limits.

Result

AWS service quotas were successfully viewed and managed using the Service Quotas service.

Conclusion

Service Quotas helps prevent overutilization of resources and enables controlled and cost-effective scaling of AWS services.